

Radiative transitions of strange heavy axial mesons into vector mesons in light-cone QCD sum rules

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(Dated: July 11, 2026)

Abstract

We calculate the radiative transitions $D_{s1}(2460) \rightarrow D_s^* \gamma$, $D_{s1}(2536) \rightarrow D_s^* \gamma$, $B_{s1}(5750) \rightarrow B_s^* \gamma$, and $B_{s1}(5830) \rightarrow B_s^* \gamma$ in light-cone QCD sum rules. The calculation is performed for the two independent axial-vector basis currents and the physical amplitudes are obtained by applying the same mixing convention as in the two-point normalization. The J_μ^B tensor-current contribution is included through the soft photon-distribution-amplitude terms and a calibrated hard-photon spectral density. With the lattice-normalized leading-twist strange-photon input and fixed $\theta = 35.3^\circ$, we find $\Gamma[D_{s1}(2460) \rightarrow D_s^* \gamma] = 92.3^{+46.4}_{-32.6}$ keV and $\Gamma[D_{s1}(2536) \rightarrow D_s^* \gamma] = 23.0^{+12.1}_{-6.5}$ keV. For the bottom-strange channels we obtain $\Gamma[B_{s1}(5750) \rightarrow B_s^* \gamma] = 10.2^{+6.0}_{-3.6}$ keV and $\Gamma[B_{s1}(5830) \rightarrow B_s^* \gamma] = 7.46^{+16.5}_{-4.16}$ keV. The charm-strange vector-final-state widths are compatible with the constructive/destructive interference pattern emphasized in recent phenomenological analyses, while the bottom-strange widths remain predictions for future radiative searches.

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I. INTRODUCTION

Radiative decays of positive-parity strange heavy mesons probe both the spin composition of the initial axial-vector state and the electromagnetic structure of the light and heavy quark lines. They are therefore a useful complement to mass spectroscopy and strong-decay measurements. The charm-strange mesons $D_{s1}(2460)$ and $D_{s1}(2536)$ are especially interesting because they are narrow and because the two physical 1^+ states are mixtures of the two axial configurations that become 3P_1 and 1P_1 in the quark-model language.

The radiative modes with a pseudoscalar final meson, $D_{s1} \rightarrow D_s \gamma$, have been studied in light-cone QCD sum rules (LCSR), quark models, and phenomenological approaches [1–8]. The vector final-state modes, $D_{s1} \rightarrow D_s^* \gamma$, are equally important because they test a different Lorentz structure and a different final-state decay constant. They also appear in the comparison tables collected by Bondar and Milstein [8], where a pronounced hierarchy between the two charm-strange axial states is explained by the relative sign of the two mixed spin components. The corresponding bottom-strange channels $B_{s1} \rightarrow B_s^* \gamma$ are experimentally unexplored and provide useful predictions for future searches.

Experimentally, $D_{s1}(2460)$ is observed in radiative and isospin-violating channels, and BaBar measured the branching fractions for $D_{s1}(2460) \rightarrow D_s \gamma$ and $D_{s1}(2460) \rightarrow D_s^* \pi^0$ [9]. There is, however, no direct partial-width measurement for $D_{s1}(2460) \rightarrow D_s^* \gamma$. The $D_{s1}(2536)$ total width is measured to be $0.92 \pm 0.03 \pm 0.04$ MeV by BaBar [10], but its radiative partial widths are not directly measured. In the bottom-strange sector, $B_{s1}(5830)$ is observed in $B^{(*)}K$ spectroscopy [11]; there is no direct information on the radiative $B_s^* \gamma$ modes.

In this work we extend the LCSR analysis to the vector-final-state transitions

$$D_{s1}(2460, 2536) \rightarrow D_s^* \gamma, \quad B_{s1}(5750), B_{s1}(5830) \rightarrow B_s^* \gamma.$$

The lower bottom-strange state is treated as a theoretical diagnostic rather than as an experimentally established resonance. We keep the strange-quark mass terms, use photon distribution amplitudes in the BBK/Rohrwild convention [12, 13], and compare the conventional susceptibility normalization with the recent lattice normalization of the leading-twist strange-photon matrix element [14]. The central predictions quoted below use the lattice-normalized input $f_{\gamma,s}^\perp(1 \text{ GeV}) = -55.1(1.9)$ MeV.

The paper is organized as follows. In Sec. II we define the currents, form factor, width formula, and basis-current sum rules. In Sec. III we give the numerical inputs, stability-window discussion, final results, no-mixing benchmark, and comparison with previous work and experiment. Section IV summarizes the conclusions.

II. THEORETICAL FRAMEWORK

We consider the transition of an axial-vector strange heavy meson X with momentum $P = p + q$ and polarization vector η_μ into a vector meson $V = D_s^*, B_s^*$, with momentum p and polarization vector ξ_ν , and a real photon with momentum q and polarization vector ε_ρ . The heavy quark is denoted by $Q = c, b$, and the light quark is s . The final vector current is

$$j_\nu^V = \bar{s}\gamma_\nu Q, \quad \langle 0 | j_\nu^V | V(p, \xi) \rangle = m_V f_V \xi_\nu. \quad (1)$$

For the initial axial-vector channel we use the two basis currents

$$J_\mu^A = \bar{s}\gamma_\mu\gamma_5 Q, \quad (2)$$

$$J_\mu^B = \frac{i}{m_Q + m_s} \bar{s}\sigma_{\mu\alpha} P^\alpha \gamma_5 Q, \quad (3)$$

where P^α is the momentum of the initial axial-vector meson. The physical-current combinations are

$$J_{1\mu} = \sin\theta J_\mu^A + \cos\theta J_\mu^B, \quad (4)$$

$$J_{2\mu} = \cos\theta J_\mu^A - \sin\theta J_\mu^B. \quad (5)$$

The lower state is coupled to $J_{1\mu}$, while the higher state is coupled to $J_{2\mu}$. Thus J_1 is assigned to $D_{s1}(2460)$ and $B_{s1}(5750)$, and J_2 to $D_{s1}(2536)$ and $B_{s1}(5830)$. The decay constants of the physical currents are defined by

$$\langle 0 | J_{i\mu} | X_i(P, \eta) \rangle = f_i m_i \eta_\mu, \quad i = 1, 2. \quad (6)$$

The $1^+ \rightarrow 1^- \gamma$ matrix element is parametrized by

$$\langle V(p, \xi) \gamma(q, \varepsilon) | X(P, \eta) \rangle = i e g^* \epsilon_{\alpha\beta\rho\sigma} \eta^\alpha \xi^{*\beta} \varepsilon^{*\rho} q^\sigma. \quad (7)$$

With this convention g^* is dimensionless. The corresponding partial width is

$$\Gamma(X \rightarrow V\gamma) = 4\pi\alpha_{\text{em}} \frac{m_X^2 - m_V^2}{16\pi m_X^3} \frac{(m_X^2 - m_V^2)^2 (m_X^2 + m_V^2)}{6m_X^2 m_V^2} |g^*|^2. \quad (8)$$

The vacuum-to-photon correlation function for a basis current is

$$\Pi_{\mu\nu}^K(p, q) = i \int d^4x e^{ip \cdot x} \langle \gamma(q, \varepsilon) | T \{ \bar{s}(x) \gamma_\nu Q(x) \bar{Q}(0) \Gamma_\mu^K s(0) \} | 0 \rangle, \quad K = A, B, \quad (9)$$

where $\Gamma_\mu^A = \gamma_\mu \gamma_5$ and $\Gamma_\mu^B = i\sigma_{\mu\alpha} P^\alpha \gamma_5 / (m_Q + m_s)$. We project the invariant amplitude multiplying

$$\epsilon_{\mu\nu\rho\sigma} \varepsilon^\rho q^\sigma. \quad (10)$$

The hadronic representation of the invariant amplitude contains the double pole

$$\Pi_K^{\text{had}}(p^2, P^2) = \frac{m_X f_K m_V f_V G_K^*}{(m_X^2 - P^2)(m_V^2 - p^2)} + \dots, \quad K = A, B, \quad (11)$$

where G_K^* denotes the basis-current form factor extracted before the physical-current rotation. In the physical basis the residue-weighted rotation gives

$$f_1 g_1^* = \sin \theta f_A G_A^* + \cos \theta f_B G_B^*, \quad (12)$$

$$f_2 g_2^* = \cos \theta f_A G_A^* - \sin \theta f_B G_B^*. \quad (13)$$

This form is used numerically; it is also the clean way to compare with the no-mixing pure-current limit.

On the QCD side the heavy-quark propagator is used in the external electromagnetic field. In momentum representation,

$$\begin{aligned} S_Q(k) = & \frac{k + m_Q}{m_Q^2 - k^2} - e_Q \int_0^1 du \left[\frac{k + m_Q}{2(m_Q^2 - k^2)^2} F^{\alpha\beta}(ux) \sigma_{\alpha\beta} + \frac{ux_\alpha F^{\alpha\beta}(ux) \gamma_\beta}{m_Q^2 - k^2} \right] \\ & - g_s \int_0^1 du \left[\frac{k + m_Q}{2(m_Q^2 - k^2)^2} G^{\alpha\beta}(ux) \sigma_{\alpha\beta} + \frac{ux_\alpha G^{\alpha\beta}(ux) \gamma_\beta}{m_Q^2 - k^2} \right] + \dots \end{aligned} \quad (14)$$

The photon emission from the light quark is represented by the standard vacuum-to-photon matrix elements of nonlocal quark and quark-gluon operators. We keep the terms proportional to m_s and use the photon distribution amplitudes summarized in Appendix A.

After the double Borel transformation, we use the symmetric choice

$$M_1^2 = M_2^2 = 2M^2, \quad u_0 = \frac{M_1^2}{M_1^2 + M_2^2} = \frac{1}{2}. \quad (15)$$

The axial basis-current form factor is obtained from the published Colangelo–De Fazio–Ozpineci spectral density and photon-DA terms [3], translated to our sign convention for χ_s . The tensor basis-current contribution is written as

$$G_B^* = G_{B,\text{soft}}^* + G_{B,\text{hard}}^*. \quad (16)$$

The soft part comes from the photon-DA projection of the tensor-current correlator. The hard part is implemented through a calibrated spectral density:

$$\rho_B^{\text{hard}}(s) = R_s^{\text{hard}}(s) \rho_{A,s}^{\text{hard}}(s) + R_Q^{\text{hard}}(s) \rho_{A,Q}^{\text{hard}}(s), \quad (17)$$

where $\rho_{A,s}^{\text{hard}}$ and $\rho_{A,Q}^{\text{hard}}$ are the strange-line and heavy-line pieces of the known axial hard density. The ratios R_s^{hard} and R_Q^{hard} are obtained from the FeynCalc triangle reduction after removing the $p \cdot q$ and $1/(p \cdot q)$ artifacts that cancel in the fully reduced axial benchmark. This construction is therefore a calibrated hard-density implementation, with the axial spectral density used as the normalization and sign check.

TABLE I. Main numerical inputs. The photon-DA normalizations are quoted at $\mu = 1$ GeV.

Parameter	Value and source
m_c, m_b, m_s	1.27(2), 4.18(3), 0.093(11) GeV [15]
$M_{D_{s1}(2460)}, M_{D_{s1}(2536)}, m_{D_s^*}$	2.4595, 2.53511, 2.1122 GeV [15]
$M_{B_{s1}(5830)}, m_{B_s^*}$	5.82870, 5.4154 GeV [11, 15]
$M_{B_{s1}(5750)}$	5.750(26) GeV, diagnostic lower state
$f_{D_s^*}, f_{B_s^*}$	0.2270(280), 0.2615(336) GeV, this work
$f_A(D_{s1}), f_T(D_{s1})$	0.225(25), 0.256(17) GeV
$f_A(B_{s1}), f_T(B_{s1})$	0.305(27), 0.285(27) GeV [16]
$\chi_s(1 \text{ GeV})$	+3.15(30) GeV ⁻² [13]
$f_{\gamma,s}^\perp(1 \text{ GeV})$	-55.1(1.9) MeV [14]

III. NUMERICAL ANALYSIS

A. Inputs and working windows

The numerical inputs are summarized in Table I. The vector decay constants $f_{D_s^*}$ and $f_{B_s^*}$ are not assumed from external models; they are calculated with a leading-order vector-current two-point baseline and assigned conservative uncertainties from the Borel-window envelope. We obtain

$$f_{D_s^*} = 0.2270 \pm 0.0280 \text{ GeV}, \quad f_{B_s^*} = 0.2615 \pm 0.0336 \text{ GeV}.$$

The central mixing angle is $\theta = 35.3^\circ$. We quote the fixed-angle result as the headline comparison because the literature table of Ref. [8] uses a fixed mixing convention; the $25^\circ \leq \theta \leq 45^\circ$ scan is kept as a systematic diagnostic.

The final windows are listed in Table II. For the bottom channels the continuum thresholds are above the pole masses by about 0.3–0.6 GeV, which is the expected location of the effective continuum in a heavy-light sum rule. The diagnostic scan gives very small three-particle fractions, good tensor-channel pole proxies, and improved width stability when the Borel interval is narrowed from $8 \leq M^2 \leq 14$ to $10 \leq M^2 \leq 14$ GeV². The axial pole proxy is moderate in the bottom sector, so we describe the windows as acceptable and stable rather than strongly pole dominated.

TABLE II. Borel and continuum-threshold windows used in the final analysis.

Channel	M^2 window	s_0 window
$D_{s1}(2460), D_{s1}(2536) \rightarrow D_s^* \gamma$	3.0–4.5 GeV ²	7.5–8.5 GeV ²
$B_{s1}(5750) \rightarrow B_s^* \gamma$	10.0–14.0 GeV ²	36.5–40.5 GeV ²
$B_{s1}(5830) \rightarrow B_s^* \gamma$	10.0–14.0 GeV ²	38.0–41.0 GeV ²

TABLE III. Final vector-final-state predictions with the lattice-normalized photon input and fixed $\theta = 35.3^\circ$.

Channel	g^*	Γ (keV)
$D_{s1}(2460) \rightarrow D_s^* \gamma$	$-1.704 [-2.083, -1.367]$	92.3 [59.7, 138.7]
$D_{s1}(2536) \rightarrow D_s^* \gamma$	$+0.654 [0.554, 0.808]$	23.0 [16.5, 35.1]
$B_{s1}(5750) \rightarrow B_s^* \gamma$	$-1.383 [-1.727, -1.138]$	10.2 [6.63, 16.24]
$B_{s1}(5830) \rightarrow B_s^* \gamma$	$+0.872 [0.583, 1.561]$	7.46 [3.30, 24.0]

B. No-mixing benchmark

Before using the mixed physical currents, we check the pure-current limit. This is the relevant comparison to same-method LCSR calculations: one sets the physical mixing aside and compares the basis-current result with the corresponding pure axial-current calculation. For $D_{s1} \rightarrow D_s^* \gamma$ we find

$$G_A^* = -0.192 \dots -0.141,$$

using Colangelo-like inputs and the original LCSR window. The published benchmark is $g^* = -0.18 \dots -0.13$ [3]. Thus the sign and normalization of the axial-current vector-final-state sum rule are reproduced. For the pseudoscalar channel, the analogous pure-current check gives $g_1 = [-0.344, -0.245]$ GeV⁻¹, to be compared with the Colangelo result $g_1 = [-0.37, -0.29]$ GeV⁻¹. This confirms that the differences in the final mixed-state predictions are not caused by a basic normalization error.

C. Final results

The final Monte Carlo results are listed in Table III. The table uses the preferred lattice photon normalization and fixed $\theta = 35.3^\circ$. The intervals are 16–84 percentile ranges.

The corresponding angle-scan medians are

$$\Gamma[D_{s1}(2460) \rightarrow D_s^* \gamma] = 93.1 \text{ keV}, \quad \Gamma[D_{s1}(2536) \rightarrow D_s^* \gamma] = 22.3 \text{ keV},$$

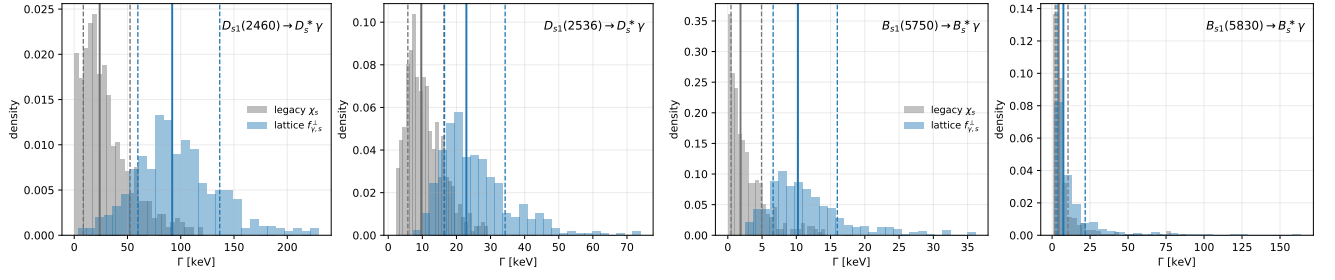


FIG. 1. Monte Carlo width distributions for the vector-final-state channels. Left: $D_{s1} \rightarrow D_s^* \gamma$. Right: $B_{s1} \rightarrow B_s^* \gamma$. The histograms compare the legacy $\chi_s \langle \bar{s}s \rangle$ normalization with the preferred lattice $f_{\gamma,s}^\perp$ normalization.

and

$$\Gamma[B_{s1}(5750) \rightarrow B_s^* \gamma] = 11.5 \text{ keV}, \quad \Gamma[B_{s1}(5830) \rightarrow B_s^* \gamma] = 2.55 \text{ keV}.$$

The larger angle sensitivity in the high bottom-strange channel comes from the same cancellation-sensitive physical-current normalization that appeared in the pseudoscalar-final-state analysis.

The tensor-current hard piece is numerically subleading but not negligible. In the charm-strange vector channel one finds

$$G_{B,\text{hard}}^* \simeq +0.16,$$

while the soft tensor-current contribution is negative and larger in magnitude. The hard term therefore partially reduces the soft tensor-current effect. The same calibrated hard-density machinery is used in the bottom sector after the replacement $c \rightarrow b$.

The uncertainty pattern differs from the pseudoscalar-final-state paper. In the old susceptibility-normalized setup the product $\chi_s \langle \bar{s}s \rangle$ remains a major normalization uncertainty. With the lattice $f_{\gamma,s}^\perp$ input, the photon normalization uncertainty is reduced. For $B_{s1}(5830) \rightarrow B_s^* \gamma$, the largest uncertainty is instead associated with the physical-current normalization, especially f_2 , because the high state is cancellation sensitive. For $B_{s1}(5750) \rightarrow B_s^* \gamma$, the dominant spread tracks the total tensor amplitude, with visible contributions from f_1 , $f_{B_s^*}$, and photon-shape parameters.

D. Comparison with literature and experiment

Table IV compares the present predictions with representative theoretical results. The final column summarizes the experimental status. It is important that the table does not list a measured partial width: direct radiative measurements of the vector-final-state modes are not available at present.

The charm-strange values show that our $D_{s1}(2460) \rightarrow D_s^* \gamma$ width is close to the large Bondar–Milstein value, while $D_{s1}(2536) \rightarrow D_s^* \gamma$ is also in the same tens-of-keV range as their estimate.

TABLE IV. Representative comparison for vector-final-state radiative widths. Widths are in keV. The bottom-row ranges collect the values of Refs. [17–19]; the other literature entries are representative values collected in Refs. [1, 2, 4–6, 8, 20].

Channel	Earlier theory	Bondar–Milstein	This work	Experimental status
$D_{s1}(2536) \rightarrow D_s^* \gamma$	2.96–22.8	29	23.0 [16.5, 35.1]	no direct radiative measurement; $\Gamma_{\text{tot}} = 0.92 \pm 0.05$ MeV
$D_{s1}(2460) \rightarrow D_s^* \gamma$	4.74–25.1	104	92.3 [59.7, 138.7]	no direct $D_s^* \gamma$ width; $D_s \gamma$ and $D_s^* \pi^0$ observed
$B_{s1}(5750) \rightarrow B_s^* \gamma$	36.9–56	20	10.2 [6.63, 16.24]	no established lower B_{s1} state or radiative measurement
$B_{s1}(5830) \rightarrow B_s^* \gamma$	53–98.8	41	7.46 [3.30, 24.0]	no direct radiative measurement; observed in $B^{(*)}K$ spectroscopy

This contrasts with the pseudoscalar-final-state channel, where the high-state amplitude is much more strongly suppressed. In our LCSR language, this difference comes from the different Lorentz structure and the relative size of the axial and tensor-current basis amplitudes after the physical-current rotation.

For the bottom-strange channels, our widths are smaller than most quark-model entries. This should not be interpreted as a direct conflict with experiment, because no radiative $B_{s1} \rightarrow B_s^* \gamma$ partial width has been measured. The established $B_{s1}(5830)$ state is observed in $B^{(*)}K$ spectroscopy [11], and the lower $B_{s1}(5750)$ row should be viewed as a diagnostic prediction for a possible lower axial configuration.

IV. CONCLUSION

We have calculated the vector-final-state radiative transitions $D_{s1}(2460, 2536) \rightarrow D_s^* \gamma$ and $B_{s1}(5750), B_{s1}(5830) \rightarrow B_s^* \gamma$ in light-cone QCD sum rules. The calculation uses the same mixed-current convention as the pseudoscalar analysis, but the final vector current, epsilon Lorentz structure, and width formula are different. The axial basis-current sum rule was checked against the Colangelo pure-current benchmark, and the tensor-current contribution was included through both soft photon-DA terms and a calibrated hard spectral density.

With the lattice-normalized strange-photon input and fixed $\theta = 35.3^\circ$, we obtain

$$\Gamma[D_{s1}(2460) \rightarrow D_s^* \gamma] = 92.3 [59.7, 138.7] \text{ keV}, \quad \Gamma[D_{s1}(2536) \rightarrow D_s^* \gamma] = 23.0 [16.5, 35.1] \text{ keV},$$

and

$$\Gamma[B_{s1}(5750) \rightarrow B_s^* \gamma] = 10.2 [6.63, 16.24] \text{ keV}, \quad \Gamma[B_{s1}(5830) \rightarrow B_s^* \gamma] = 7.46 [3.30, 24.0] \text{ keV}.$$

The $D_s^* \gamma$ results are broadly consistent with the Bondar–Milstein interference picture, especially for the large $D_{s1}(2460)$ vector-channel width. In the bottom-strange sector the predicted widths

are smaller than several quark-model estimates, and direct radiative measurements are needed to distinguish among the calculations.

The comparison with experiment is presently qualitative. There are no direct partial-width measurements for the vector-final-state radiative modes. The BaBar and CMS results provide important spectroscopy and total-width context, but not a direct radiative benchmark. The predictions given here can therefore serve as targets for future searches of $D_{s1}(2536) \rightarrow D_s^* \gamma$ and $B_{s1}(5830) \rightarrow B_s^* \gamma$, and as tests of the spin-mixing structure of strange heavy axial mesons.

Appendix A: Photon DAs and calibrated hard density

The photon distribution amplitudes are used in the BBK/Rohrwild notation [12, 13]. For the leading-twist two-particle DA we use

$$\phi_\gamma(u) = 6u(1-u). \quad (\text{A1})$$

The conventional normalization is expressed through $\chi_s \langle \bar{s}s \rangle$, while the preferred normalization replaces this product by the lattice-determined strange photon coupling $f_{\gamma,s}^\perp(1 \text{ GeV}) = -55.1(1.9) \text{ MeV}$ [14]. In both normalizations the same two- and three-particle DA shapes are used.

The hard tensor-current density is obtained by calibrating the FeynCalc triangle reduction to the known axial hard density. The axial density is split into strange-emission and heavy-emission pieces,

$$\rho_A^{\text{hard}} = \rho_{A,s}^{\text{hard}} + \rho_{A,Q}^{\text{hard}}.$$

The tensor density is then reconstructed line by line as in Eq. (17). This removes the $p \cdot q$ and $1/(p \cdot q)$ artifacts that cancel in the fully reduced axial expression and preserves the known logarithmic spectral density as the normalization benchmark.

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